

The Normal Distribution

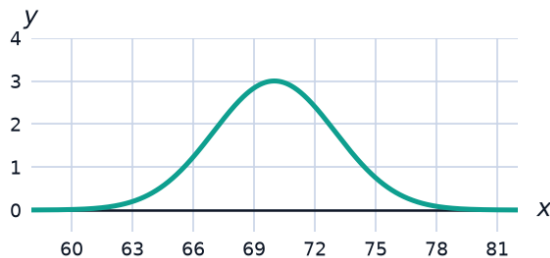


Properties of the normal distribution

- 1 State three key properties of a normal distribution's shape.
 - 2 Explain what the two parameters μ and σ control in a normal distribution's shape.
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The 68-95-99.7 rule

- 3 Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.



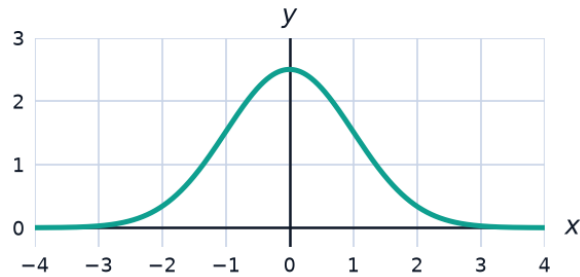
- 4 Using the same distribution, find the approximate percentage of adult men taller than 76 inches.
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Standardizing and using the normal distribution

- 5 Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.
 - 6 Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.
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Visualizing the standard normal distribution

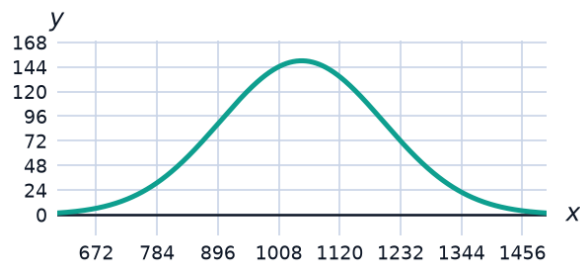
- 7 The graph shows the standard normal distribution ($\mu = 0, \sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).



- a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.
- b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$.

Working backward: finding values from percentiles

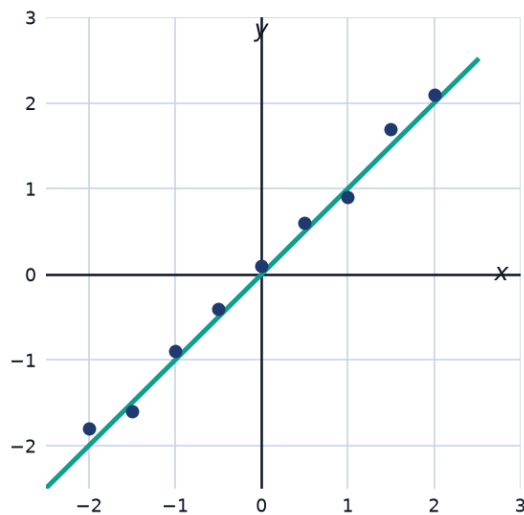
- 8 SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.



- 9 Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

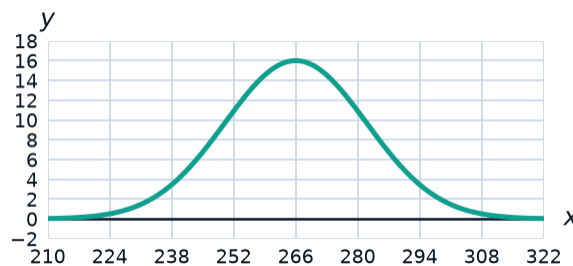
Assessing normality

- 10 Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.



Applying the normal distribution to real data

- 11 Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z-score for a pregnancy lasting 250 days.

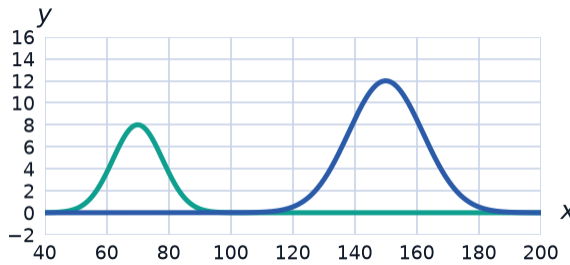


- 12 Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.

Combining two independent normal random variables

- 13 A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T . (This is a simplified illustrative combination, not a real physical unit.)
- 14 Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

- 15** A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.
- 16** Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.



Multiple choice — easy 15

- 17** A normal distribution has mean 64 and standard deviation 9. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (46, 82)
 - b. (37, 91)
 - c. (46, 81)
 - d. (47, 82)
- 18** A normal distribution has mean 162 and standard deviation 13. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.
- a. (136, 188)
 - b. (149, 174)
 - c. (149, 175)
 - d. (150, 175)

- 19** A normal distribution has mean 332 and standard deviation 16. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.
- a. (268, 396)
 - b. (284, 379)
 - c. (285, 380)
 - d. (284, 380)
- 20** A normal distribution has mean 228 and standard deviation 4. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.
- a. (225, 232)
 - b. (220, 236)
 - c. (224, 232)
 - d. (224, 231)
- 21** A normal distribution has mean 457 and standard deviation 11. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.
- a. (425, 490)
 - b. (424, 490)
 - c. (424, 489)
 - d. (413, 501)
- 22** A normal distribution has mean 195 and standard deviation 12. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.
- a. (159, 230)
 - b. (160, 231)
 - c. (159, 231)
 - d. (147, 243)

- 23** A normal distribution has mean 50 and standard deviation 14. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.
- a. (22, 78)
 - b. (36, 64)
 - c. (36, 63)
 - d. (37, 64)
- 24** A normal distribution has mean 288 and standard deviation 6. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.
- a. (283, 294)
 - b. (282, 294)
 - c. (276, 300)
 - d. (282, 293)
- 25** A normal distribution has mean 438 and standard deviation 35. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (368, 508)
 - b. (368, 507)
 - c. (333, 543)
 - d. (369, 508)
- 26** A normal distribution has mean 138 and standard deviation 40. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (58, 218)
 - b. (18, 258)
 - c. (58, 217)
 - d. (59, 218)

- 27** A normal distribution has mean 481 and standard deviation 20. Using the 68-95-99.7 rule, find the range containing the middle 99.7% of values.
- a. (421, 541)
 - b. (422, 541)
 - c. (421, 540)
 - d. (401, 561)
- 28** A normal distribution has mean 88 and standard deviation 18. Using the 68-95-99.7 rule, find the range containing the middle 68% of values.
- a. (70, 106)
 - b. (70, 105)
 - c. (52, 124)
 - d. (71, 106)
- 29** A normal distribution has mean 64 and standard deviation 15. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (35, 94)
 - b. (19, 109)
 - c. (34, 93)
 - d. (34, 94)
- 30** A normal distribution has mean 361 and standard deviation 35. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (291, 430)
 - b. (256, 466)
 - c. (292, 431)
 - d. (291, 431)

- 31 A normal distribution has mean 414 and standard deviation 13. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.
- a. (388, 439)
 - b. (389, 440)
 - c. (375, 453)
 - d. (388, 440)
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Multiple choice — medium 15

- 32 A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.
- a. 69.15%
 - b. 72.15%
 - c. 66.15%
 - d. 30.85%
- 33 A normal distribution has mean $\mu = 94$ and standard deviation $\sigma = 2$. Find the approximate percentage of values below $x = 89$, to two decimal places.
- a. 99.38%
 - b. 2.62%
 - c. 3.62%
 - d. 0.62%

- 34 A normal distribution has mean $\mu = 365$ and standard deviation $\sigma = 18$. Find the approximate percentage of values below $x = 356$, to two decimal places.
- a. 30.85%
 - b. 27.85%
 - c. 69.15%
 - d. 33.85%
- 35 A normal distribution has mean $\mu = 64$ and standard deviation $\sigma = 10$. Find the approximate percentage of values below $x = 39$, to two decimal places.
- a. 0.62%
 - b. 2.62%
 - c. 3.62%
 - d. 99.38%
- 36 A normal distribution has mean $\mu = 160$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 184$, to two decimal places.
- a. 90.32%
 - b. 96.32%
 - c. 93.32%
 - d. 6.68%
- 37 A normal distribution has mean $\mu = 477$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 527$, to two decimal places.
- a. 96.38%
 - b. 0.62%
 - c. 97.38%
 - d. 99.38%

- 38 A normal distribution has mean $\mu = 485$ and standard deviation $\sigma = 10$. Find the approximate percentage of values below $x = 505$, to two decimal places.
- a. 94.72%
 - b. 2.28%
 - c. 95.72%
 - d. 97.72%
- 39 A normal distribution has mean $\mu = 174$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 198$, to two decimal places.
- a. 6.68%
 - b. 90.32%
 - c. 93.32%
 - d. 96.32%
- 40 A normal distribution has mean $\mu = 110$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 126$, to two decimal places.
- a. 15.87%
 - b. 81.13%
 - c. 87.13%
 - d. 84.13%
- 41 A normal distribution has mean $\mu = 454$ and standard deviation $\sigma = 6$. Find the approximate percentage of values below $x = 457$, to two decimal places.
- a. 72.15%
 - b. 69.15%
 - c. 66.15%
 - d. 30.85%

- 42 A normal distribution has mean $\mu = 338$ and standard deviation $\sigma = 20$. Find the approximate percentage of values below $x = 388$, to two decimal places.
- a. 99.38%
 - b. 96.38%
 - c. 0.62%
 - d. 97.38%
- 43 A normal distribution has mean $\mu = 268$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 280$, to two decimal places.
- a. 96.87%
 - b. 99.87%
 - c. 97.87%
 - d. 0.13%
- 44 A normal distribution has mean $\mu = 99$ and standard deviation $\sigma = 2$. Find the approximate percentage of values below $x = 104$, to two decimal places.
- a. 99.38%
 - b. 97.38%
 - c. 96.38%
 - d. 0.62%
- 45 A normal distribution has mean $\mu = 318$ and standard deviation $\sigma = 16$. Find the approximate percentage of values below $x = 342$, to two decimal places.
- a. 90.32%
 - b. 6.68%
 - c. 96.32%
 - d. 93.32%

- 46 A normal distribution has mean $\mu = 343$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 335$, to two decimal places.
- a. 5.28%
 - b. 2.28%
 - c. 4.28%
 - d. 97.72%
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Multiple choice — hard 15

- 47 A normal distribution has mean $\mu = 144$ and standard deviation $\sigma = 16$. Find the approximate percentage of values between $x_1 = 136$ and $x_2 = 152$, to two decimal places.
- a. 69.15%
 - b. 34.29%
 - c. 42.29%
 - d. 38.29%
- 48 A normal distribution has mean $\mu = 237$ and standard deviation $\sigma = 2$. Find the approximate percentage of values between $x_1 = 233$ and $x_2 = 240$, to two decimal places.
- a. 91.04%
 - b. 87.04%
 - c. 95.04%
 - d. 93.32%

- 49 A normal distribution has mean $\mu = 57$ and standard deviation $\sigma = 4$. Find the approximate percentage of values between $x_1 = 63$ and $x_2 = 69$, to two decimal places.
- a. 10.55%
 - b. 2.55%
 - c. 99.87%
 - d. 6.55%
- 50 A normal distribution has mean $\mu = 354$ and standard deviation $\sigma = 18$. Find the approximate percentage of values between $x_1 = 318$ and $x_2 = 363$, to two decimal places.
- a. 66.87%
 - b. 69.15%
 - c. 70.87%
 - d. 62.87%
- 51 A normal distribution has mean $\mu = 123$ and standard deviation $\sigma = 10$. Find the approximate percentage of values between $x_1 = 133$ and $x_2 = 148$, to two decimal places.
- a. 11.24%
 - b. 19.24%
 - c. 99.38%
 - d. 15.24%
- 52 Scores on an exam are approximately normal with mean $\mu = 390$ and standard deviation $\sigma = 15$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.
- a. 365
 - b. 405
 - c. 425
 - d. 415

- 53 Scores on an exam are approximately normal with mean $\mu = 837$ and standard deviation $\sigma = 50$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.
- a. 909
 - b. 755
 - c. 919
 - d. 929
- 54 Scores on an exam are approximately normal with mean $\mu = 399$ and standard deviation $\sigma = 10$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.
- a. 405
 - b. 415
 - c. 425
 - d. 383
- 55 Scores on an exam are approximately normal with mean $\mu = 295$ and standard deviation $\sigma = 20$. A student scores at the 10th percentile, corresponding to $z \approx -1.28$. Find that student's approximate score.
- a. 279
 - b. 259
 - c. 321
 - d. 269
- 56 Scores on an exam are approximately normal with mean $\mu = 762$ and standard deviation $\sigma = 50$. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.
- a. 826
 - b. 836
 - c. 698
 - d. 816

- 57 A company manufactures parts with a measurement approximately normal, mean 152 and standard deviation 5. A part is considered defective if its measurement is below 142 or above 162 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 0.3%
 - b. 5%
 - c. 32%
 - d. 2.5%
- 58 A company manufactures parts with a measurement approximately normal, mean 224 and standard deviation 27. A part is considered defective if its measurement is below 170 or above 278 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 32%
 - b. 2.5%
 - c. 5%
 - d. 0.3%
- 59 A company manufactures parts with a measurement approximately normal, mean 85 and standard deviation 28. A part is considered defective if its measurement is below 1 or above 169 (i.e. outside $\mu \pm 3\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 0.3%
 - b. 0.15%
 - c. 5%
 - d. 32%

- 60** A company manufactures parts with a measurement approximately normal, mean 84 and standard deviation 15. A part is considered defective if its measurement is below 39 or above 129 (i.e. outside $\mu \pm 3\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 0.3%
 - b. 5%
 - c. 0.15%
 - d. 32%
- 61** A company manufactures parts with a measurement approximately normal, mean 22 and standard deviation 30. A part is considered defective if its measurement is below -8 or above 52 (i.e. outside $\mu \pm 1\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.
- a. 32%
 - b. 16%
 - c. 0.3%
 - d. 5%